Example 1

Consider the oxidation of nitric oxide to form NO_2

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$$

(a). Express the rate of the reaction in terms of changes in the concentration of NO, O_2 and NO_2.

(b). At a particular instant, when $[O_2]$ is decreasing at 0.2 mol $L^{-1}s^{-1}$ at what rate is $[NO_2]$ increasing at that instant?

Solution:

a) $$\text{Rate} = \frac{-1}{2}\frac{d[NO]}{dt} = \frac{-d[O_2]}{dt} = \frac{1}{2}\frac{d[NO_2]}{dt}$$

b) $$\frac{-d[O_2]}{dt} = \frac{1}{2}\frac{d[NO_2]}{dt}$$

$$\frac{d[NO_2]}{dt} = 2x\left(\frac{-d[O_2]}{dt}\right) = 2 \text{ x}0.2 \text{ mol } L^{-1}s^{-1}$$

$$= 0.4 \text{ mol } L^{-1}s^{-1}$$

Evaluate yourself 1

1). Write the rate expression for the following reactions, assuming them as elementary reactions.

i) $3A + 5B_2 \longrightarrow 4CD$

ii) $X_2 + Y_2 \longrightarrow 2XY$

2). Consider the decomposition of $N_2O_5(g)$ to form $NO_2(g)$ and $O_2(g)$. At a particular instant N_2O_5 disappears at a rate of $2.5x10^{-2}$ mol $dm^{-3}s^{-1}$. At what rates are NO_2 and O_2 formed? What is the rate of the reaction?

Example 2

1. What is the order with respect to each of the reactant and overall order of the following reactions?

(a). $5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(l) + 3H_2O(l)$

The experimental rate law is

$\text{Rate} = k\,[Br^-][BrO_3^-][H^+]^2$

(b). $CH_3CHO(g) \xrightarrow{\Delta} CH_4(g) + CO(g)$

the experimental rate law is

$\text{Rate} = k\,[CH_3CHO]^{\frac{3}{2}}$

Solution:

a) First order with respect to Br^-, first order with respect to BrO_3^- and second order with respect to H^+. Hence the overall order of the reaction is equal to $1 + 1 + 2 = 4$

b) Order of the reaction with respect to acetaldehyde is $\frac{3}{2}$ and overall order is also $\frac{3}{2}$

Example 3

2. The rate of the reaction $x + 2y \rightarrow$ product is $4 \text{ x } 10^{-3}$ mol $L^{-1}s^{-1}$, if $[x]=[y]=0.2$ M and rate constant at 400K is $2 \text{ x } 10^{-2}s^{-1}$, What is the overall order of the reaction.

Solution :

$\text{Rate} = k\,[x]^n[y]^m$

$4 \text{ x } 10^{-3} \text{ mol } L^{-1}s^{-1} = 2 \text{ x } 10^{-2}s^{-1}(0.2\text{mol } L^{-1})^n(0.2\text{mol } L^{-1})^m$

$$\frac{4 \text{ x } 10^{-3} \text{ mol } L^{-1}s^{-1}}{2 \text{ x } 10^{-2}s^{-1}} = (0.2)^{n+m}\left(\text{mol } L^{-1}\right)^{n+m}$$

$$0.2\left(\text{mol } L^{-1}\right) = (0.2)^{n+m}\left(\text{mol } L^{-1}\right)^{n+m}$$

Comparing the powers on both sides

The overall order of the reaction $n + m = 1$

Evaluate yourself 2

1). For a reaction, $X + Y \longrightarrow$ product ; quadrupling [x] , increases the rate by a

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